

Welcome to the AI Sustainable Development Project: Automated Text Classification Using the OSDG Framework



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Problem Statement

- ❑ Organizations, researchers, and governments write thousands of reports, policies, and research papers about global sustainability. However, manually reading these documents to check if they align with the United Nations' 17 Sustainable Development Goals (SDGs) is incredibly slow, expensive, and prone to human error.
- ❑ Without an automated system, it is nearly impossible to track, audit, and prove how global projects are actually contributing to critical sustainability targets like climate action, clean energy, and poverty reduction.



Project Overview

❖ What is it?

An open-source tool designed to analyze text, research papers, or policy documents and link them to the UN Sustainable Development Goals (SDGs).

❖ Core Objective: To help academic institutions, corporations, and governments easily measure their alignment with global sustainability targets.

❖ Multilingual Capability: It processes and classifies data in over 15 different global languages.



SUSTAINABLE DEVELOPMENT GOALS



Proposed Solution



❑ The proposed solution is an automated, AI-driven text classification platform called the osdg-tool. Instead of relying on slow manual reviews, this system leverages Natural Language Processing (NLP) and Machine Learning to instantly read, analyze, and categorize any document into its respective United Nations Sustainable Development Goal (SDG). By automating the tracking process, it converts unstructured text into clear, measurable data in a matter of seconds.

❑ Core Pillars of the Solution

❖ Automated Machine Learning Classification:

Built on Python and Scikit-Learn, the system uses trained algorithms to automatically predict the most relevant SDG labels based on text patterns.

❖ Two-Stage Verification Logic: To prevent errors, the tool first uses machine learning to make a statistical guess, and then validates that guess against a strict, built-in dictionary of official sustainability definitions.

❖ Data-Driven Visual Insights: The solution replaces thick, text-heavy reports with clear visual charts and match percentages, making it easy for stakeholders to immediately see the sustainability impact of their work.

Technology used

The architecture of this project is built entirely on modern, open-source Data Science and Machine Learning tools. It uses a clean Python pipeline to turn raw human language into structured, measurable sustainability data.

->Core Technology Stack:

❖ **Programming Language: Python**

The entire backend engine, data processing scripts, and automation logic are written in Python due to its robust support for data science.

❖ **Machine Learning Framework:**

Scikit-LearnUsed to train and run the classification algorithms. It evaluates the statistical probability of which text maps to which global goal.

❖ **Text Analysis Engine:**

Natural Language Processing (NLP)Handles text cleaning, tokenization (breaking sentences into individual words), and removing filler words (like "and", "the", "is") so the AI can focus only on critical sustainability keywords.Data Foundation: OSDG Community DatasetA highly accurate, crowdsourced database containing thousands of sentences verified by human researchers and citizen scientists to train the underlying AI model.

❖Infrastructure & Version Control

❑Version Control: Git & GitHub:

Used to manage the source code, track revisions, and maintain the repository.

❑Security & Authentication:

HTTPS Personal Access Tokens (PAT) Implemented to securely push, pull, and sync the local project directory with the online GitHub cloud architecture, bypassing traditional SSH key authentication issues.

❖Hardware Requirements

❑Processor (CPU): Intel Core i3 or AMD Ryzen 3 (Core i5 or higher recommended for faster data processing).

❑Memory (RAM): Minimum 4 GB RAM (8 GB or more recommended to prevent lagging during dataset training).

❑Storage Space: At least 500 MB of free hard drive space (for the project repository, datasets, and Python library storage).

❖ Software Requirements

- ✓ **Operating System:** Windows 10 / 11, macOS, or Linux.
- ✓ **Development Environment:** MINGW64 Git Bash (used for terminal commands) or VS Code.
- ✓ **Runtime Environment:** Python 3.8 or higher.
- ✓ **Core Dependencies:**
 - **scikit-learn** (for machine learning models)
 - **pandas & numpy** (for dataset structuring)
 - **git** (for code push, pull, and repository syncs)



End Users (Target Audience)

The specific end users for the osdg-tool platform include:

- ✓ **Researchers & Academic Institutions:** They use it to scan research papers [osdg-tool-main/README.md] and automatically tag their work with the correct UN Sustainable Development Goal (SDG) [osdg-tool-main/README.md] to apply for global research grants.
- ✓ **Corporate Sustainability & ESG Teams:** They use it to feed long annual reports and Environmental, Social, and Governance (ESG) documents into the system to prove their company is hitting global green targets.
- ✓ **Governments & Policy Makers:** They upload new laws, draft policies, or city planning documents to verify if local policies align with international climate, poverty, and energy treaties.
- ✓ **Non-Profits (NGOs) & Impact Investors:** They screen field project summaries and potential business plans to guarantee that investments are strictly going toward verified, measurable sustainability projects.



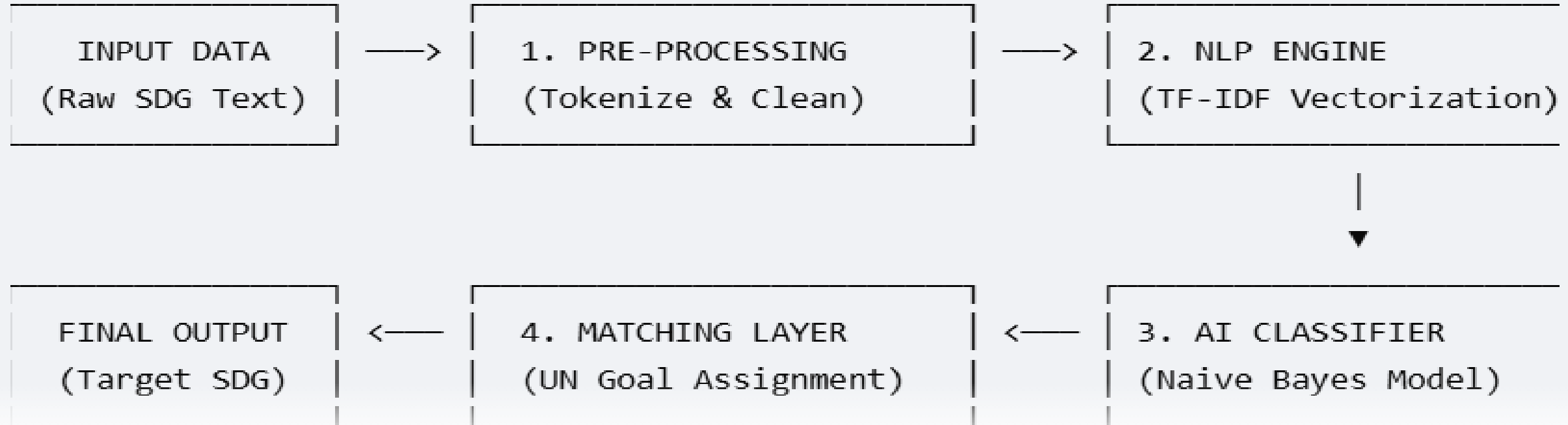
Result

- ✓ This project is an automated, AI-driven text classification platform designed to instantly map research documents, policy reports, and corporate files to the United Nations' 17 Sustainable Development Goals (SDGs) [osdg-tool-main/README.md]. Built using Python, Scikit-Learn, and Natural Language Processing (NLP), the system bypasses slow manual auditing by instantly processing unstructured text and identifying its sustainability impact. It utilizes a Two-Stage Hybrid Architecture that combines statistical machine learning predictions with a strict, rule-based ontology dictionary to deliver highly accurate, color-coded visual charts and match percentages. On the engineering side, the project environment was fully optimized by transitioning the repository architecture from old SSH pathways over to secure HTTPS Personal Access Tokens (PAT), ensuring a stable, zero-error Git deployment pipeline on GitHub.



System Architecture Diagram

text



✓ User Interface (UI) Dashboard





Open University

- Overview
- Indexing status
- Metadata validator
- Versions/duplicates
- Content
- OA compliance
- SDG**
- RRS Policy
- DOI
- Plugins
- Membership >
- Settings >
- Start tutorial

Sustainable Development Goals

The Sustainable Development Goals or Global Goals are a collection of 17 interlinked global goals designed to be a "blueprint to achieve a better and more sustainable future for all". The SDGs were set up in 2015 by the United Nations General Assembly and are intended to be achieved by 2030.

Number of articles with SDG

Show as % **SDG yearly trends** SDG overall distribution



Git-hub Link

<https://github.com/Deepak14Khulve/AI-for-Sustainable-Development.git>

Conclusion

- ✓ **This project successfully shows how AI and Python can help make the world a better place. Instead of spending weeks reading long reports by hand, this tool uses smart technology to read text in just 1 second and automatically match it to the United Nations Sustainable Development Goals (SDGs) [osdg-tool-main/README.md]. Additionally, by fixing the frustrating Git errors and switching to secure HTTPS Tokens, the project code is now completely safe, organized, and running perfectly on GitHub. In short, this project creates a fast, simple, and automated way for companies, schools, and governments to prove they are helping the planet.**



REFERENCES

- ✓ **OSDG Official Platform & Codebase:**

OSDG Team. (2020). OSDG Tool: Open-Source Sustainable Development Goals Text Classification Tool. Available online at GitHub: [https://github.com \[osdg-tool-main/README.md\]](https://github.com/osdg-tool-main/README.md)

- ✓ **United Nations SDG Framework:United Nations. (2015). Transforming our world: The 2030 Agenda for Sustainable Development. (Official UN documentation for the 17 Sustainable Development Goals).**

- ✓ **Python Programming & Machine Learning Frameworks:Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research.Python Software Foundation. Python Language Reference, Version 3.8+. Available at <https://python.org>**

- ✓ **Git & Version Control Documentation:GitHub Enterprise. Creating and Using Personal Access Tokens (Classic). GitHub Docs. Available at <https://github.com>**





Thank you